

QUANTUM COMPUTING

ITS TIME IS NOW (WK1-PT1)

Professor Nella Ludlow, PhD

Wright State University

Dayton, OH

Quantum Computing The Time is Now

Forbes

Billionaires

Innovation

Leadership

Money

Consumer

Industry

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Why Quantum Computing's Time Is Now



Nella Ludlow Contributor
COGNITIVE WORLD Contributor Group @

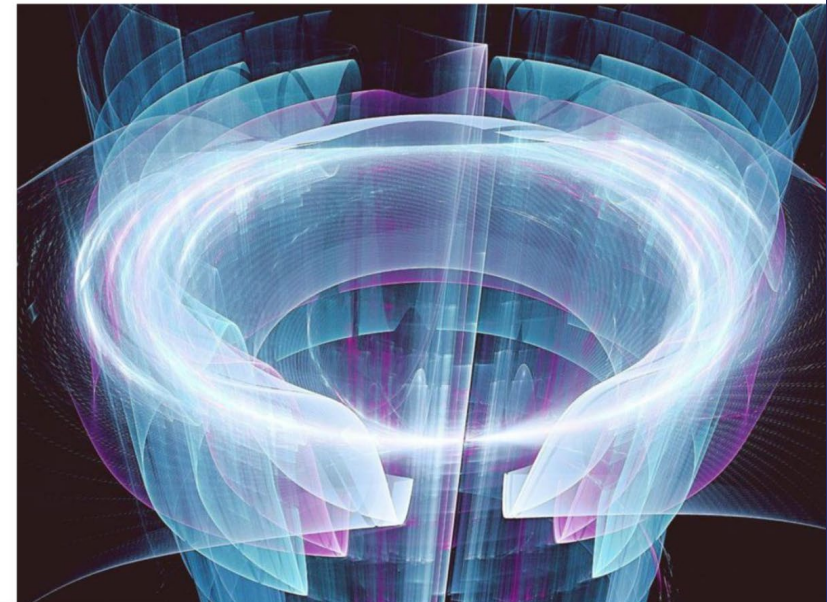
AI & Big Data

Writer on Technology, AI, Big Data, and Quantum Computing

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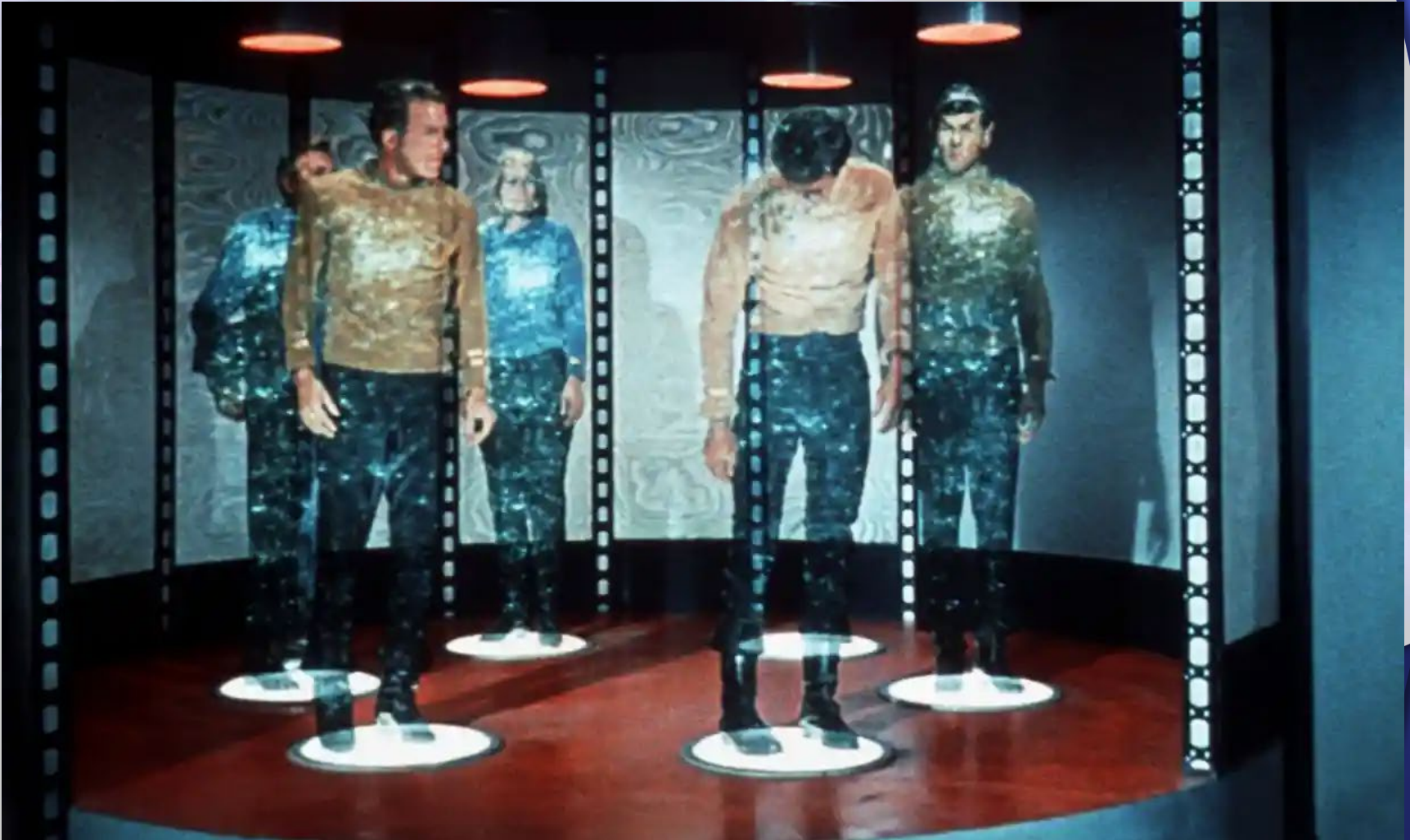


UBER'S SECRET GOLD MINE • FINTECH'S 50 HOTTEST



24 Feb 2019

“Beam Me Up, Scotty!”





QUANTUM LEAP

Amazon.com: Finish - Quantum - x

amazon.com/Finish-Technology-Dishwasher-Detergent-Dishwashing/dp/B0857XP96L/ref=asc_df_B0857XP96L/?tag=hyprod-20&linkCode=df0&hvadid=459642201277&hvpos=&hvnetw=g&hvra...

Deliver to Nella Port Town... 98368 Health, Household & Baby Care

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Finish - Quantum - 64ct - Dishwasher Detergent - Powerball - Ultimate Clean & Shine - Dishwashing Tablets - Dish Tabs

Size: 64 Count (Pack of 1)



Roll over image to zoom in

on food

- Quick dissolving dishwasher Power Gel washes away residue for the ultimate clean & shine
- Advanced powder with bleach cleans tough stains like tea & coffee
- Contains Glass Protect Action to help prevent glass corrosion & redensit

Deliver every: 3 months (Most common)

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Understanding Quantum Computing



WSJ Interview with Bill Gates and Satya Nadella

<https://www.wsj.com/articles/a-rare-joint-interview-with-microsoft-ceo-satya-nadella-and-bill-gates-1506358852>

WSJ: Can you explain Quantum Computing in one sentence, so my 72-year old mother could understand?

Bill Gates laughed and then mused, "I know a lot of physics and a lot of math. But the one place where they put up slides and it is hieroglyphics, it's quantum."

Then the WSJ turned to Satya Nadella and asked him if he could explain Quantum Computing, he responded, "I don't think so. I wish I could."

While Gates and Nadella were perplexed to define quantum computing, Nadella added, "we think of [quantum computing] as our next-generation cloud."

RICHARD FEYNMAN

FIRST PROPOSED QUANTUM COMPUTATION

“because nature isn't classical, dammit”

EDITED BY TONY HEY AND ROBIN W. ALLEN

RICHARD P. FEYNMAN

FEYNMAN LECTURES ON COMPUTATION



Simulating Physics with Computers

Richard P. Feynman

Department of Physics, California Institute of Technology, Pasadena, California 91107

Received May 7, 1981

4. QUANTUM COMPUTERS—UNIVERSAL QUANTUM SIMULATORS

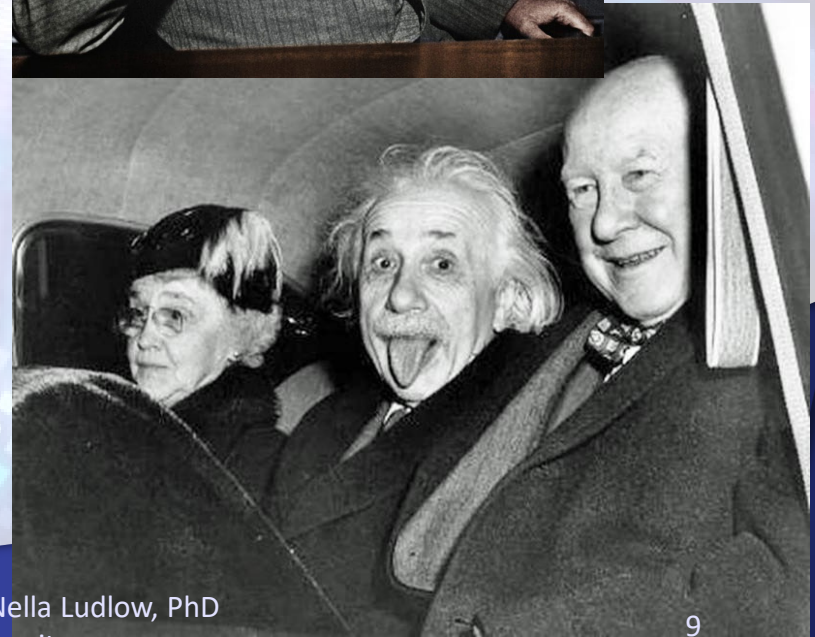
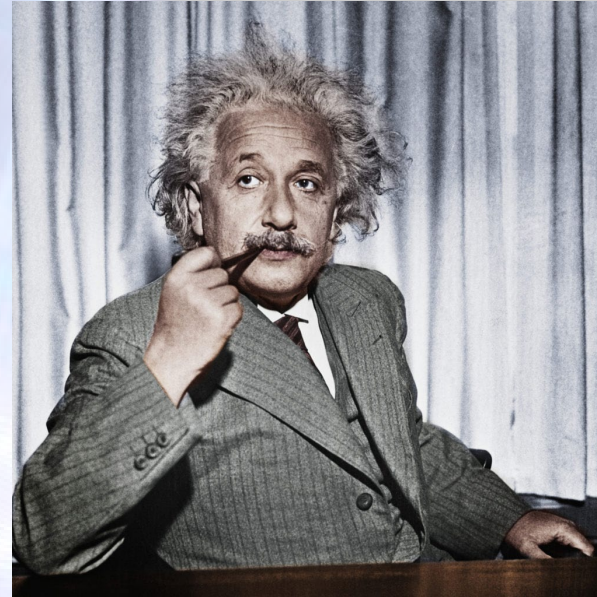
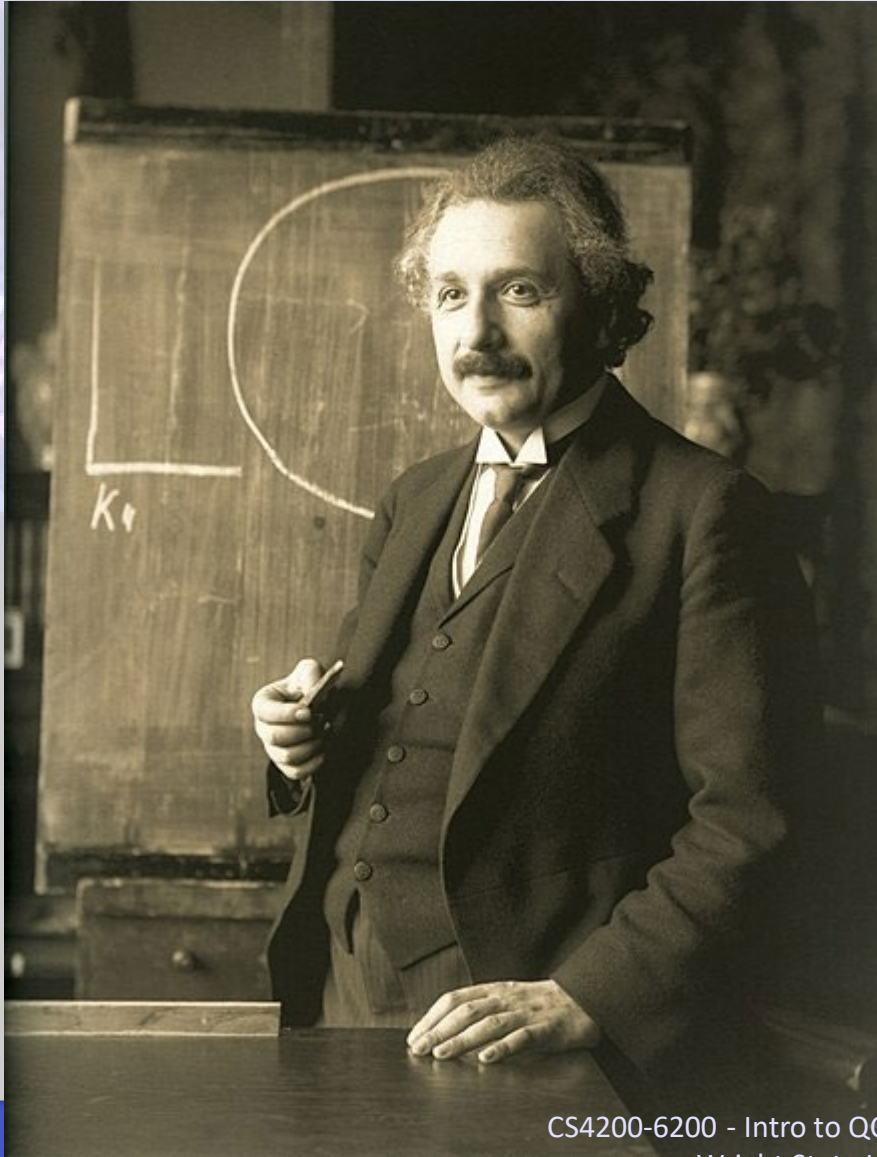
The first branch, one you might call a side-remark, is, Can you do it with a new kind of computer—a quantum computer? (I'll come back to the other branch in a moment.) Now it turns out, as far as I can tell, that you can simulate this with a quantum system, with quantum computer elements. It's not a Turing machine, but a machine of a different kind. If we disregard

And I'm not happy with all the analyses that go with just the classical theory, because nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical, and by golly it's a wonderful problem, because it doesn't look so easy. Thank you.

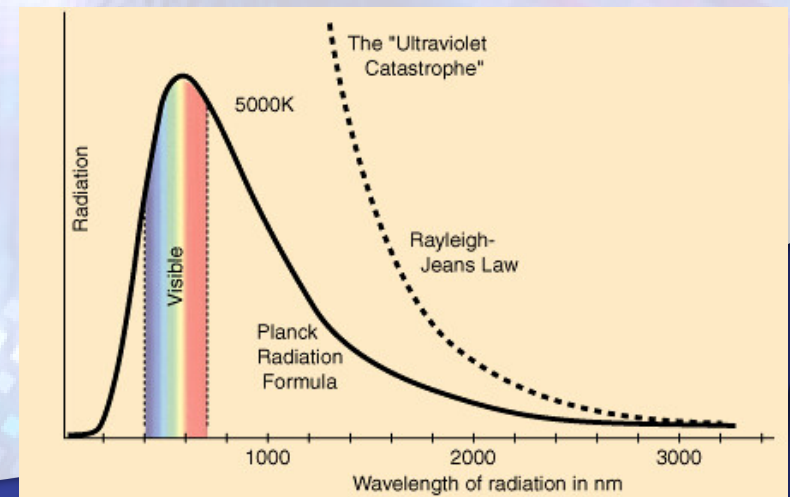
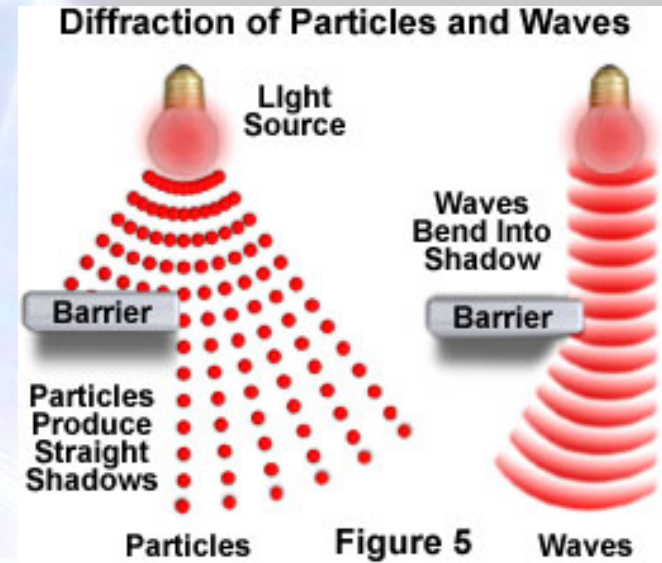
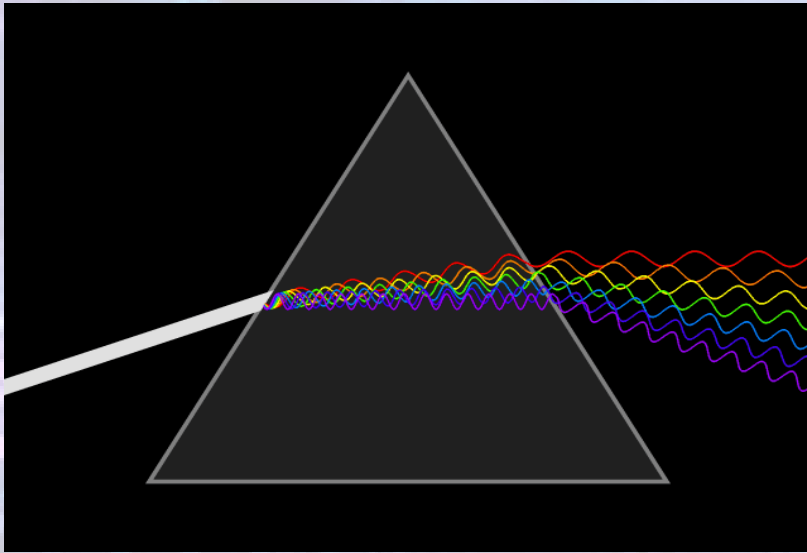
ALGORITHMS AND SPEED



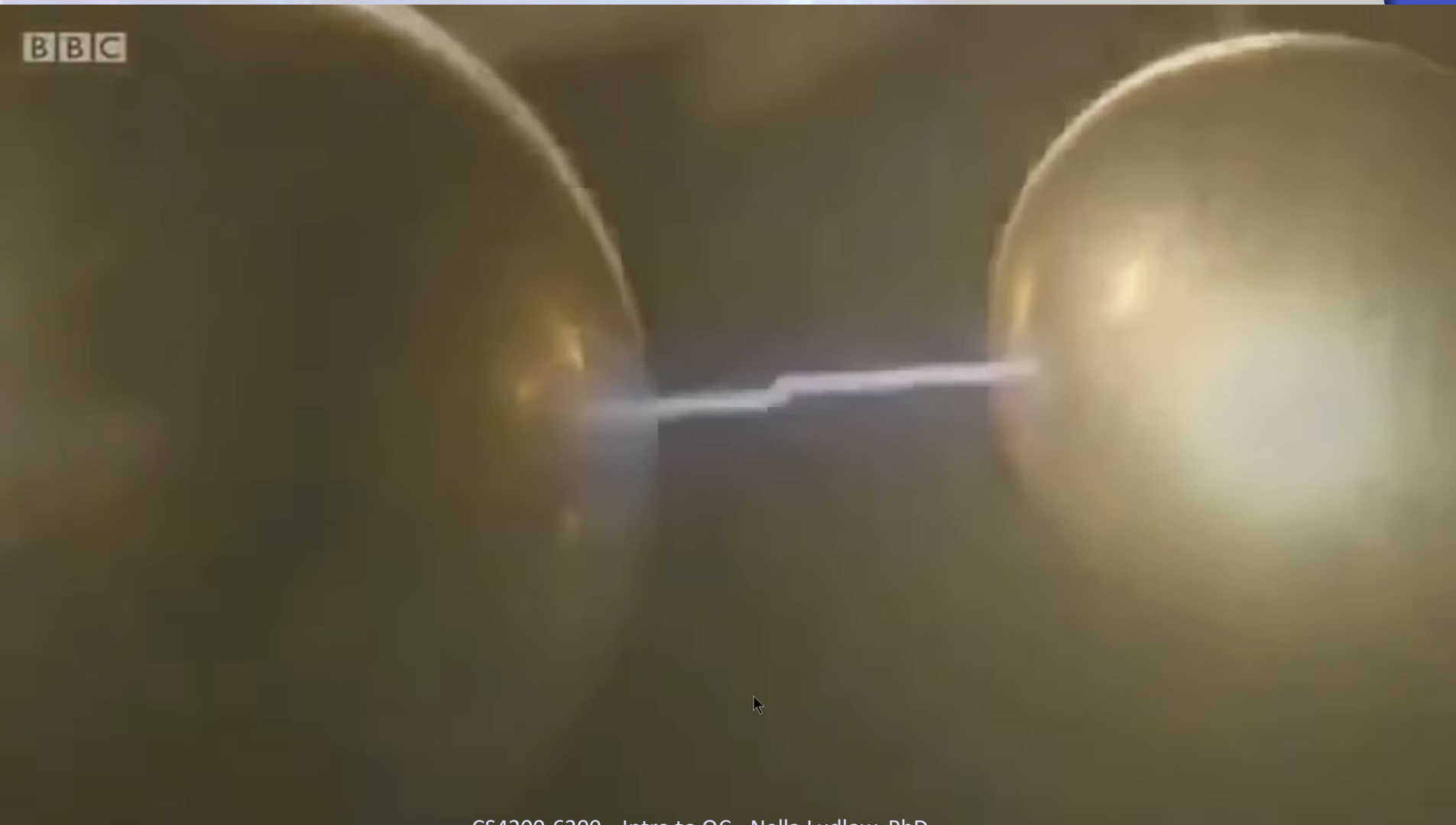
LOOK AT PHYSICS



Light as a Wave



LIGHT AS A WAVE

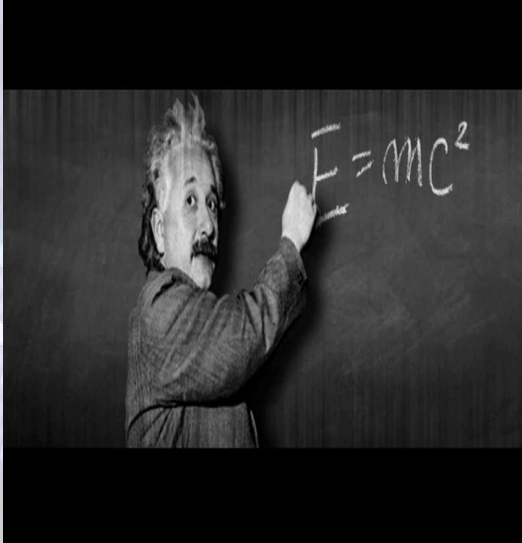


LIGHT AS A PARTICLE



QUANTUM

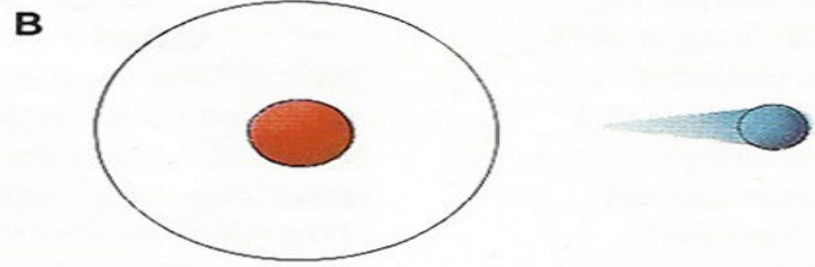
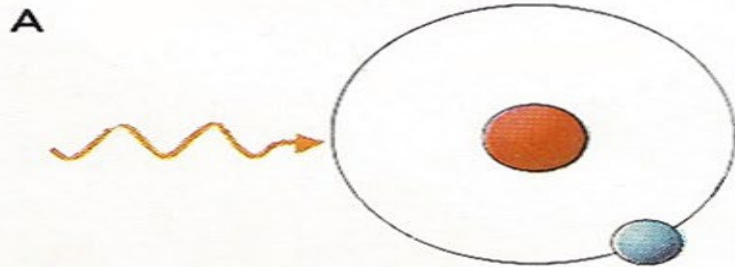
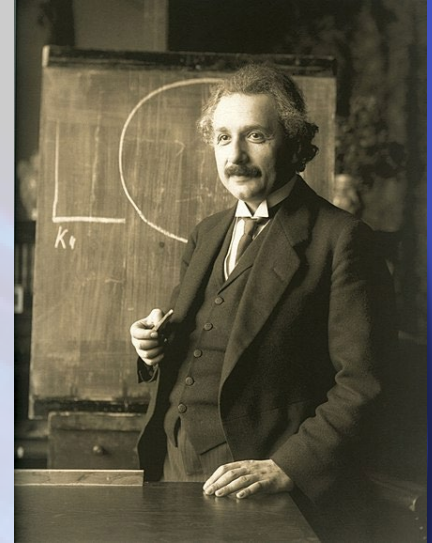
Light as a Particle



“WHAT ARE
LIGHT QUANTA?

OF COURSE TODAY EVERY
RASCAL THINKS HE KNOWS
THE ANSWER, BUT HE IS DELUDING HIMSELF.”

-Albert Einstein



The photoelectric effect was explained by Einstein in 1905 as the absorption of a quantum of energy [A] by an atom and the resulting emission of an electron [B].